

MOOYEON KIM

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SUMMARY

IoT and embedded systems engineer experienced in developing sensor-actuator based two-tier architecture applications using microcontrollers and networking protocols.

EDUCATION

Northeastern University , Boston, MA	Expected 2027
MS in Cyber-Physical Systems, GPA 4.0	
Purdue University , West Lafayette, IN	2022
BA in Electrical and Computer Engineering	

PROJECTS

Connected Device Project	Spring 2026
- Designing and implementing an end-to-end IoT solution as part of a semester project - Will integrate constrained devices, gateway logic, and cloud services to build a complete device-to-cloud data pipeline. - Will define outcomes, use cases, and core requirements, then develop functionality spanning embedded firmware, network communication, and cloud integration.	
Temperature Monitoring System	Fall 2025
- Designed and implemented a real-time temperature monitoring system using ESP32 and BMP280 sensor with device-side data processing in C/C++ - Enabled real-time data transmission to users through Wi-Fi - Integrated hardware, firmware, and user-facing applications into an IoT pipeline	
Smart Hard Hat	Spring 2022
- Led software development for a smart hard hat integrating LoRa technology for real-time communication - Implemented a fall detection feature using accelerometer sensor data - Developed a user interface to display real-time sensor information for monitoring hazards	

SKILLS

C, C++, C#, Python, MIPS, Verilog, Java, Networking Protocol, WireShark, MATLAB
Linux, Microprocessor Systems, Unity Engine, HTML

WORK EXPERIENCE

Jeodong Middle-School , Goyang, South Korea	07/2022 - 05/2024
Administrative & Facility Support Assistant	
- Assisted in facility management and repairs, ensuring efficient resolution of technical issues - Managed library resources and inventory, implementing an organized system to enhance accessibility and minimize shortages	

LEADERSHIP & ACTIVITIES

Purdue Baseball Team	2023 – 2024
Purdue Korean Association (PKA) - Event Organizer	2021 – 2022